AI-Based Smart Renewable Energy Asset Performance Monitoring System

CO1 AT2

Agile sprint Planning

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1. Executive Summary

The global imperative for sustainable energy has led to a rapid expansion of renewable energy infrastructure, including solar farms and wind turbines. While these assets offer immense potential for clean power generation, their optimal performance is contingent upon continuous, intelligent monitoring and proactive maintenance. Traditional approaches often fall short, leading to reduced efficiency, increased operational costs, and premature equipment degradation. This document outlines the Agile Sprint Planning and User Story Workshop for an **AI-Based Smart Renewable Energy Asset Performance Monitoring System**. This innovative platform aims to leverage artificial intelligence (AI), Internet of Things (IoT), drone technology, and cloud computing to provide real-time insights into equipment performance, environmental conditions, and maintenance requirements. By predicting equipment failures, optimizing maintenance schedules, and recommending performance improvements, the system will maximize energy generation, minimize downtime, and significantly enhance the operational efficiency and sustainability of renewable energy assets. This report details the project's vision, defines key stakeholders, outlines epics and user stories, establishes acceptance criteria, and presents a prioritized product backlog, culminating in a focused sprint plan for initial development.

2. Project Vision and Stakeholders

Project Vision Statement

To establish the **AI-Based Smart Renewable Energy Asset Performance Monitoring System** as the leading intelligent platform that empowers renewable energy providers to achieve unparalleled operational efficiency, maximize energy generation, and ensure the long-term sustainability and profitability of their assets through predictive analytics and optimized maintenance strategies.

Project Objectives

The development of this AI-based monitoring system is guided by a set of clear objectives designed to address the core challenges faced by renewable energy asset owners and operators. These objectives are critical for ensuring the system delivers tangible value and meets the overarching vision.

1. **Continuous Monitoring of Renewable Energy Assets:** Implement a robust data ingestion pipeline capable of collecting real-time operational data from diverse sources, including solar panels, wind turbines, and battery systems. This objective ensures a comprehensive and up-to-date view of asset health and performance. The system will integrate with existing SCADA (Supervisory Control and Data Acquisition) systems and IoT sensors to capture granular data points such as power output, temperature, vibration, and environmental parameters [1].

2. **Predict Equipment Failures Using AI:** Develop and deploy advanced AI and machine learning models to analyze real-time and historical data for anomaly detection and predictive failure analysis. This includes identifying patterns indicative of impending equipment malfunctions, such as bearing degradation in wind turbines or inverter failures in solar arrays. The AI models will utilize techniques such as time-series analysis, deep

learning (e.g., LSTMs), and anomaly detection algorithms to provide early warnings, thereby preventing catastrophic failures and enabling proactive interventions [2].

3. **Optimize Maintenance Scheduling:** Automate the generation of optimized maintenance schedules based on AI-driven predictions, asset criticality, and resource availability. This objective aims to shift from reactive or time-based maintenance to a predictive approach, reducing unnecessary inspections and prioritizing critical repairs. The system will consider factors such as weather forecasts, spare parts inventory, and technician availability to create efficient maintenance plans that minimize downtime and operational costs [3].

4. **Analyze Weather Impacts on Energy Generation:** Integrate real-time and forecasted weather data to assess its impact on energy production and to provide context for performance anomalies. This includes correlating solar irradiance, wind speed, temperature, and precipitation with actual energy output to differentiate between weather-induced fluctuations and equipment underperformance. This analysis will enhance the accuracy of performance benchmarks and inform operational adjustments [4].

5. **Improve Asset Utilization:** Provide actionable recommendations and insights to enhance the operational efficiency and lifespan of renewable energy assets. This objective focuses on identifying opportunities for performance optimization, such as adjusting turbine pitch angles, cleaning solar panels, or optimizing battery charge/discharge cycles based on predicted demand and market prices. The system will leverage prescriptive analytics to suggest the best course of action for maximizing energy capture and asset longevity.

6. **Generate Energy Performance Reports:** Develop customizable dashboards and reporting tools that provide comprehensive insights into energy generation, asset health, maintenance activities, and financial performance. These reports will cater to various stakeholders, from plant operators to executive management, offering clear visualizations and key performance indicators (KPIs) to track progress against operational and financial goals. Reports will include historical trends, real-time status, and future projections.

7. **Reduce Operational Downtime:** By enabling predictive maintenance and optimized scheduling, the system will significantly reduce unplanned outages and the duration of planned maintenance activities. This objective directly contributes to increased energy availability and revenue generation. The platform will facilitate rapid fault diagnosis and resolution, minimizing the time assets are offline.

8. **Support Sustainable Energy Production:** Contribute to the broader goal of sustainable energy by ensuring renewable assets operate at their peak, thereby maximizing their contribution to the energy grid and reducing reliance on fossil fuels. The system's ability to optimize performance and extend asset life directly supports environmental stewardship and the transition to a greener energy future.

Stakeholder Register

Effective development and deployment of the AI-Based Smart Renewable Energy Asset Performance Monitoring System require a clear understanding of all involved stakeholders, their interests, and their potential impact on the project. The following table identifies key stakeholders and their primary concerns:

Stakeholder Group | Primary Interests

3. Epics and User Stories

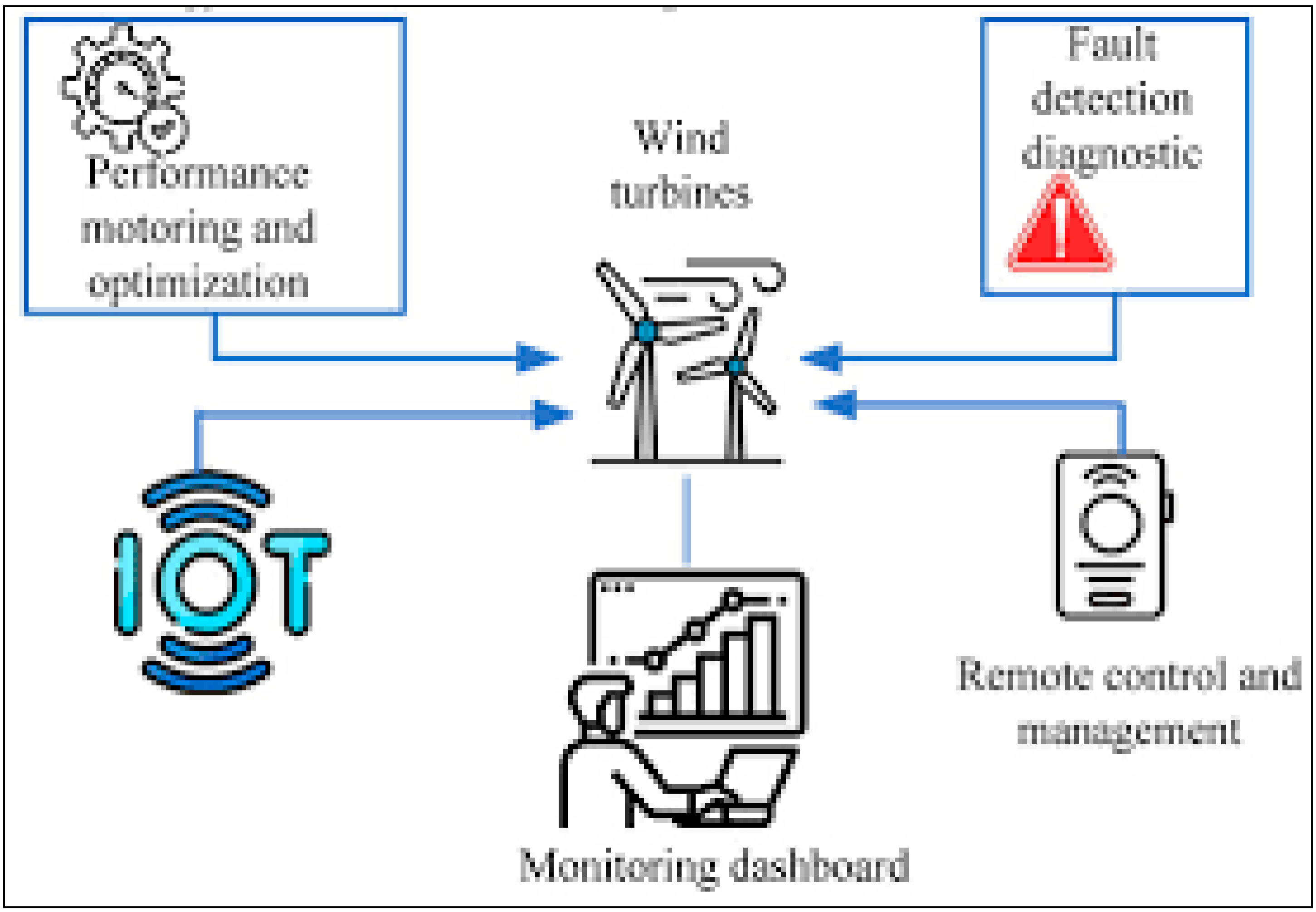
To effectively manage the development of the AI-Based Smart Renewable Energy Asset Performance Monitoring System, requirements are broken down into Epics and further refined into detailed User Stories. Epics represent large bodies of work that can be broken down into several smaller user stories, providing a high-level view of the system's functionality. User stories, following the format "As a , I want , so that ," describe specific functionalities from the perspective of an end-user, ensuring that development is user-centric and value-driven.

Epics

The following epics categorize the major functionalities required for the monitoring system:

- **Epic 1: Multi-Source Data Ingestion & Normalization**
  - **Description:** This epic focuses on establishing a robust and scalable data pipeline capable of collecting, processing, and normalizing diverse data streams from various renewable energy assets and external sources. This includes data from IoT sensors on solar panels and wind turbines, SCADA systems, battery management systems, and external weather APIs. The goal is to ensure data quality, consistency, and availability for subsequent AI analysis.
- **Epic 2: AI Predictive Analytics Engine**
  - **Description:** This epic encompasses the development and integration of advanced AI and machine learning models to analyze the ingested data. The primary objective is to predict equipment failures, detect anomalies in real-time, and identify patterns indicative of performance degradation. This engine will be the core intelligence of the system, providing early warnings and actionable insights for proactive maintenance.
- **Epic 3: Performance Optimization & Weather Analysis**
  - **Description:** This epic is dedicated to leveraging AI and weather data to optimize the performance of renewable energy assets. It involves analyzing the impact of environmental conditions on energy generation, identifying opportunities for operational adjustments (e.g., turbine pitch optimization, solar panel cleaning schedules), and providing recommendations to maximize energy output and efficiency.
- **Epic 4: Maintenance Workflow Automation**
  - **Description:** This epic focuses on automating and streamlining the maintenance process based on the insights generated by the AI engine. It includes features for generating optimized maintenance schedules, creating and managing work orders, integrating with field service management tools, and potentially incorporating drone-based inspection data for detailed fault assessment.
- **Epic 5: Visualization & Executive Reporting**

- **Description:** This epic addresses the need for intuitive and comprehensive data visualization and reporting capabilities. It involves developing customizable dashboards that present real-time asset performance, predictive insights, maintenance status, and key performance indicators (KPIs) to various stakeholders, from plant operators to executive management. The goal is to provide clear, actionable information for informed decision-making.



User Story Catalog

Below is a selection of detailed user stories, categorized by their respective epics, illustrating the specific functionalities required by different users of the system. These stories adhere to the "As a , I want , so that " format.

Epic 1: Multi-Source Data Ingestion & Normalization

1. **As a Plant Manager**, I want to **automatically ingest real-time sensor data from all solar panels**, so that I can **monitor their operational status continuously**.
2. **As an O&M Engineer**, I want to **integrate SCADA data from wind turbines**, so that I can **access comprehensive operational parameters for analysis**.
3. **As a Data Scientist**, I want to **access normalized and cleaned data from all asset types**, so that I can **develop and train accurate AI models**.
4. **As a System Administrator**, I want to **configure new data sources (e.g., new weather APIs)**, so that I can **expand the system's data collection capabilities without manual coding**.

Epic 2: AI Predictive Analytics Engine

5. **As an O&M Engineer**, I want to **receive alerts for predicted wind turbine gearbox failures**, so that I can **schedule proactive maintenance before a critical breakdown occurs** [5].
6. **As a Plant Manager**, I want to **see a risk score for each solar inverter**, so that I can **prioritize inspections and replacements based on potential failure impact**.
7. **As a Data Scientist**, I want to **retrain AI models with new historical data**, so that I can **improve the accuracy of failure predictions over time**.
8. **As an O&M Engineer**, I want to **identify abnormal vibration patterns in wind turbine blades**, so that I can **detect early signs of structural damage**.

Epic 3: Performance Optimization & Weather Analysis

9. **As a Plant Manager**, I want to **view the impact of current weather conditions on energy generation forecasts**, so that I can **understand expected output variations**.
10. **As an O&M Engineer**, I want to **receive recommendations for optimal solar panel cleaning schedules based on soiling levels and weather forecasts**, so that I can **maximize energy capture**.
11. **As a Data Scientist**, I want to **analyze the correlation between wind speed and turbine efficiency**, so that I can **identify optimal operating parameters**.
12. **As a Grid Operator**, I want to **access real-time energy output predictions adjusted for weather**, so that I can **better manage grid stability and supply-demand balance**.

Epic 4: Maintenance Workflow Automation

13. **As an O&M Engineer**, I want to **automatically generate a work order when a critical failure is predicted**, so that I can **initiate the repair process without delay**.

14. **As a Maintenance Scheduler**, I want to **optimize technician routes and assignments based on predicted failures and technician skill sets**, so that I can **reduce travel time and improve response efficiency**.
15. **As a Drone Operator**, I want to **upload drone inspection imagery and have it automatically analyzed for blade damage**, so that I can **quickly assess the extent of damage and plan repairs**.

Epic 5: Visualization & Executive Reporting

16. **As a Plant Manager**, I want to **view a real-time dashboard of overall plant performance and asset health**, so that I can **quickly identify underperforming assets**.
17. **As an Executive**, I want to **receive monthly reports on energy generation, operational costs, and ROI**, so that I can **assess the financial performance of our renewable assets**.
18. **As an O&M Engineer**, I want to **access detailed historical performance data for a specific asset**, so that I can **troubleshoot recurring issues and analyze trends**.
19. **As a Compliance Officer**, I want to **generate audit-ready reports on system uptime and environmental impact**, so that I can **meet regulatory requirements**.
20. **As a Data Analyst**, I want to **export raw and processed data for custom analysis**, so that I can **conduct deeper investigations and generate specialized reports**.

4. Acceptance Criteria

Acceptance criteria define the conditions that a user story must satisfy to be considered complete and functional. They provide clear, testable statements that ensure the developed features meet the user's needs and expectations. For this project, acceptance criteria will primarily follow the **Given-When-Then** format, which is a behavior-driven development (BDD) approach that describes a scenario, the action taken, and the expected outcome. This format enhances clarity, reduces ambiguity, and facilitates automated testing.

Given-When-Then Scenarios for Key User Stories

Below are detailed acceptance criteria for a selection of key user stories, demonstrating the expected behavior and outcomes of the system.

User Story 5: As an O&M Engineer, I want to receive alerts for predicted wind turbine gearbox failures, so that I can schedule proactive maintenance before a critical breakdown occurs.

- **Scenario 1: Gearbox Failure Prediction and Alert Generation**
  - **Given** the AI predictive analytics engine has processed real-time vibration, temperature, and oil analysis data from a wind turbine gearbox for the past 72 hours,
  - **And** the AI model predicts a high probability (e.g., >85%) of gearbox failure within the next 30 days,
  - **When** the prediction threshold is met,
  - **Then** the system shall generate a critical alert notification to the assigned O&M Engineer via email and in-app notification,

- **And** the alert shall include the turbine ID, predicted failure component (gearbox), confidence score, and estimated time to failure.

- **Scenario 2: Alert Escalation for Unacknowledged Predictions**
  - **Given** a critical alert for a predicted gearbox failure has been sent to the O&M Engineer,
  - **And** the alert remains unacknowledged for more than 4 hours during operational hours (8 AM - 5 PM local time),
  - **When** the unacknowledged period expires,
  - **Then** the system shall escalate the alert to the Plant Manager via email and in-app notification,
  - **And** the escalation notification shall include all details of the original alert and indicate that it was unacknowledged.

User Story 6: As a Plant Manager, I want to see a risk score for each solar inverter, so that I can prioritize inspections and replacements based on potential failure impact.

- **Scenario 1: Inverter Risk Score Calculation and Display**
  - **Given** the AI system has analyzed historical performance data, temperature logs, and error codes for all solar inverters,
  - **And** the system has calculated a dynamic risk score (e.g., 1-10, where 10 is highest risk) for each inverter based on its health and predicted degradation,
  - **When** the Plant Manager accesses the solar farm dashboard,
  - **Then** the dashboard shall display a list of all inverters with their current risk scores, sorted by highest risk first,
  - **And** clicking on an inverter shall show a detailed view of its risk factors and historical performance.
- **Scenario 2: Risk Score Update on New Data**
  - **Given** new operational data (e.g., increased temperature, new error codes) is ingested for a specific solar inverter,
  - **When** the AI system re-evaluates the inverter's health,
  - **Then** the inverter's risk score shall be updated within 15 minutes of data ingestion,
  - **And** if the risk score crosses a predefined critical threshold (e.g., >7), an alert shall be generated to the Plant Manager.

User Story 10: As an O&M Engineer, I want to receive recommendations for optimal solar panel cleaning schedules based on soiling levels and weather forecasts, so that I can maximize energy capture.

- **Scenario 1: Cleaning Recommendation Generation**
  - **Given** the system has analyzed current soiling levels (e.g., via drone imagery or sensor data) for a section of solar panels,
  - **And** the system has access to a 7-day weather forecast (e.g., no rain expected, high dust probability),
  - **And** the predicted energy loss due to soiling exceeds a configurable threshold (e.g., 2%),

- **When** the daily optimization routine runs,
- **Then** the system shall recommend a cleaning schedule for the affected panel section, including optimal date and time,
- **And** the recommendation shall be accessible via the O&M Engineer's dashboard and email.

- **Scenario 2: Recommendation Adjustment for Adverse Weather**
  - **Given** a cleaning recommendation has been issued for tomorrow,
  - **And** the weather forecast is updated to predict heavy rain on the recommended cleaning day,
  - **When** the system re-evaluates the schedule,
  - **Then** the system shall automatically postpone the cleaning recommendation to the next optimal day without rain,
  - **And** notify the O&M Engineer of the schedule change.

User Story 13: As an O&M Engineer, I want to automatically generate a work order when a critical failure is predicted, so that I can initiate the repair process without delay.

- **Scenario 1: Automatic Work Order Creation**
  - **Given** a critical alert for a predicted equipment failure (e.g., wind turbine bearing) has been confirmed by the O&M Engineer,
  - **And** the system has access to predefined work order templates for the specific failure type,
  - **When** the O&M Engineer marks the alert as 'actioned' or 'approved',
  - **Then** the system shall automatically generate a new work order in the integrated CMMS (Computerized Maintenance Management System),
  - **And** the work order shall include the asset ID, predicted failure, recommended actions, and a link to the original alert.
- **Scenario 2: Work Order Status Update**
  - **Given** a work order has been automatically generated for a predicted failure,
  - **And** a technician updates the status of the work order to 'In Progress' in the CMMS,
  - **When** the CMMS system sends a status update to the monitoring platform,
  - **Then** the monitoring platform shall reflect the updated work order status on the asset's dashboard,
  - **And** notify relevant stakeholders (e.g., Plant Manager) of the progress.

Technical Constraints and Non-Functional Requirements

In addition to functional acceptance criteria, the system must also adhere to several non-functional requirements and technical constraints to ensure its reliability, performance, and security.

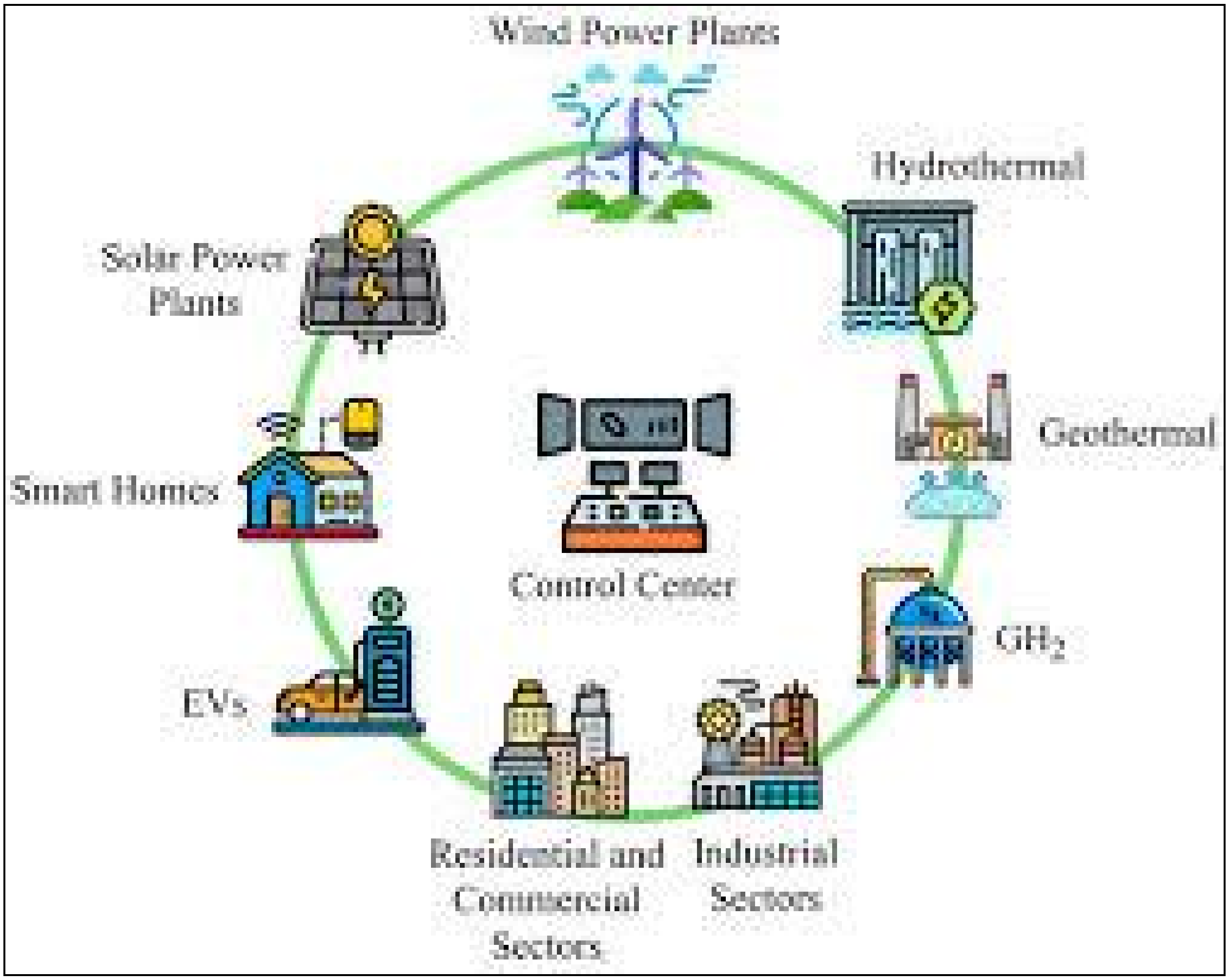
- **Performance:**
  - Data ingestion latency for real-time sensor data shall not exceed 5 seconds.

- AI model inference for failure prediction shall complete within 10 seconds of receiving new data.
- Dashboard data refresh rate shall be configurable, with a minimum of 60 seconds.
- The system shall support monitoring of at least 10,000 assets concurrently without performance degradation.

- **Scalability:**
  - The platform architecture shall be cloud-native and designed for horizontal scalability to accommodate future growth in asset count and data volume.
  - The data storage solution shall be capable of handling petabytes of time-series data.
- **Reliability & Availability:**
  - The system shall maintain an uptime of 99.9% (excluding scheduled maintenance).
  - Data redundancy and backup mechanisms shall be in place to prevent data loss.
  - Disaster recovery procedures shall ensure data and service restoration within 4 hours.
- **Security:**
  - All data in transit and at rest shall be encrypted using industry-standard protocols (e.g., TLS 1.2+, AES-256).
  - User authentication shall implement multi-factor authentication (MFA).
  - Role-based access control (RBAC) shall be enforced to limit user access to authorized functionalities and data.
  - Regular security audits and penetration testing shall be conducted.
- **Maintainability:**
  - The codebase shall adhere to clean code principles and be well-documented.
  - Automated testing (unit, integration, end-to-end) shall cover at least 80% of the codebase.
  - Deployment processes shall be automated using CI/CD pipelines.
- **Usability:**
  - The user interface shall be intuitive and responsive across various devices (desktop, tablet).
  - Dashboards and reports shall be customizable by end-users.
  - Comprehensive user documentation and training materials shall be provided.

5. Prioritized Product Backlog

The product backlog for the AI-Based Smart Renewable Energy Asset Performance Monitoring System is a dynamic, ordered list of all known features, functions, requirements, enhancements, and fixes that are needed for the product. It is continuously refined and prioritized based on business value, customer



needs, technical feasibility, and project dependencies. For this project, we will employ the **MoSCoW prioritization technique** to categorize backlog items and a **Business Value vs. Technical Complexity matrix** to guide prioritization.

MoSCoW Prioritization

The MoSCoW method categorizes requirements into four types:

- **Must Have:** These are critical requirements that are fundamental to the project's success and must be included in the current delivery. Without them, the product would be unusable or illegal.
- **Should Have:** These are important requirements that add significant value but are not essential for the initial release. Their absence would be noticeable but would not prevent the system from functioning.
- **Could Have:** These are desirable requirements that would improve the user experience or add convenience but are not necessary. They are often included if time and resources permit.

- **Won't Have:** These are requirements that are explicitly excluded from the current delivery. They may be considered for future releases.

Business Value vs. Technical Complexity Mapping

To further refine prioritization, each user story will be assessed based on its perceived business value and technical complexity. This helps in identifying "quick wins" (high value, low complexity) and "major projects" (high value, high complexity).

Prioritized Product Backlog Table

The following table presents a prioritized list of user stories, incorporating MoSCoW categorization, business value, technical complexity, and dependencies.

ID	User Story	Epic	MoSCoW	Business Value	Technical Complexity	Dependencies
US-01	As a Plant Manager, I want to automatically ingest real-time sensor data from all solar panels, so that I can monitor their operational status continuously.	Epic 1	Must Have	High	Medium	None
US-02	As an O&M Engineer, I want to integrate SCADA data	Epic 1	Must Have	High	Medium	None

ID	User Story	Epic	MoSCoW	Business Value	Technical Complexity	Dependencies
	from wind turbines, so that I can access comprehensive operational parameters for analysis.					
US-03	As a Data Scientist, I want to access normalized and cleaned data from all asset types, so that I can develop and train accurate AI models.	Epic 1	Must Have	High	High	US-01, US-02
US-05	As an O&M Engineer, I want to receive alerts for predicted wind turbine gearbox	Epic 2	Must Have	High	High	US-03

ID	User Story	Epic	MoSCoW	Business Value	Technical Complexity	Dependencies
	failures, so that I can schedule proactive maintenance before a critical breakdown occurs.					
US-16	As a Plant Manager, I want to view a real-time dashboard of overall plant performance and asset health, so that I can quickly identify underperforming assets.	Epic 5	Must Have	High	Medium	US-01, US-02
US-06	As a Plant Manager, I want to see a risk score for each solar inverter,	Epic 2	Should Have	High	Medium	US-03

ID	User Story	Epic	MoSCoW	Business Value	Technical Complexity	Dependencies
	so that I can prioritize inspections and replacements based on potential failure impact.					
US-09	As a Plant Manager, I want to view the impact of current weather conditions on energy generation forecasts, so that I can understand expected output variations.	Epic 3	Should Have	Medium	Medium	US-01, US-02
US-13	As an O&M Engineer, I want to automatically generate a work	Epic 4	Should Have	High	Medium	US-05

ID	User Story	Epic	MoSCoW	Business Value	Technical Complexity	Dependencies
	order when a critical failure is predicted, so that I can initiate the repair process without delay.					
US-17	As an Executive, I want to receive monthly reports on energy generation, operational costs, and ROI, so that I can assess the financial performance of our renewable assets.	Epic 5	Should Have	Medium	Low	US-16
US-10	As an O&M Engineer, I want to receive	Epic 3	Could Have	Medium	High	US-03, US-09

ID	User Story	Epic	MoSCoW	Business Value	Technical Complexity	Dependencies
	recommendations for optimal solar panel cleaning schedules based on soiling levels and weather forecasts, so that I can maximize energy capture.					
US-14	As a Maintenance Scheduler, I want to optimize technician routes and assignments based on predicted failures and technician skill sets, so that I can reduce travel	Epic 4	Could Have	Medium	High	US-13

ID	User Story	Epic	MoSCoW	Business Value	Technical Complexity	Dependencies
	time and improve response efficiency.					
US-15	As a Drone Operator, I want to upload drone inspection imagery and have it automatically analyzed for blade damage, so that I can quickly assess the extent of damage and plan repairs.	Epic 4	Won't Have (Sprint 1)	High	High	None

6. Sprint Backlog (Sprint 1)

The Sprint Backlog is a subset of the Product Backlog selected for implementation during the upcoming sprint. For Sprint 1, the focus is on establishing the foundational data ingestion pipeline and creating a basic visualization dashboard to demonstrate early value. This sprint will lay the groundwork for subsequent AI model integration and advanced features.

Selected User Stories for Sprint 1

The following user stories have been selected for Sprint 1 based on their high priority (Must Have) and foundational nature:

1. **US-01:** As a Plant Manager, I want to automatically ingest real-time sensor data from all solar panels, so that I can monitor their operational status continuously.
2. **US-02:** As an O&M Engineer, I want to integrate SCADA data from wind turbines, so that I can access comprehensive operational parameters for analysis.
3. **US-16:** As a Plant Manager, I want to view a real-time dashboard of overall plant performance and asset health, so that I can quickly identify underperforming assets.

Task Breakdown

To ensure manageable and trackable work, each selected user story is broken down into actionable development tasks.

US-01: Ingest real-time sensor data from solar panels

- **Task 1.1 (Backend):** Design and implement API endpoints for receiving data from solar panel IoT sensors.
- **Task 1.2 (Data Engineering):** Develop a data pipeline to validate, clean, and store incoming sensor data in the time-series database.
- **Task 1.3 (DevOps):** Set up secure communication channels (e.g., MQTT, HTTPS) between sensors and the ingestion API.
- **Task 1.4 (Testing):** Create automated tests to verify data ingestion rates and data integrity.

US-02: Integrate SCADA data from wind turbines

- **Task 2.1 (Backend):** Develop connectors to interface with existing wind turbine SCADA systems (e.g., OPC UA, Modbus).
- **Task 2.2 (Data Engineering):** Map SCADA data points to the standardized data model and implement data transformation logic.
- **Task 2.3 (DevOps):** Configure secure network access and authentication for SCADA integration.
- **Task 2.4 (Testing):** Perform integration testing with simulated SCADA data to ensure accurate data mapping and storage.

US-16: View real-time dashboard of overall plant performance

- **Task 16.1 (Frontend):** Design the user interface for the main dashboard, including layout and widget placement.
- **Task 16.2 (Frontend):** Implement data visualization components (e.g., line charts, gauges) to display aggregated power output and asset status.
- **Task 16.3 (Backend):** Develop API endpoints to serve aggregated performance data to the frontend dashboard.
- **Task 16.4 (Testing):** Conduct user acceptance testing (UAT) to ensure the dashboard meets usability and performance requirements.

7. Effort Estimation (Story Points)

Effort estimation is a crucial part of sprint planning, helping the team understand the capacity required to complete the selected work. We will use **Story Points** as the unit of measure, employing the **Fibonacci sequence** (1, 2, 3, 5, 8, 13, 21) to represent relative effort, complexity, and uncertainty. The estimation process involves the entire development team, often using techniques like Planning Poker to reach a consensus.

Estimation Table and Justification

The following table presents the estimated story points for the user stories selected for Sprint 1, along with a justification for the assigned value.

ID	User Story	Story Points	Justification
US-01	Ingest real-time sensor data from solar panels	5	The effort involves setting up new API endpoints and data pipelines. While the technology is understood, the volume of data and the need for robust validation add complexity. It is a foundational task requiring careful design.
US-02	Integrate SCADA data from wind turbines	8	Integrating with legacy SCADA systems is often more complex than handling modern IoT sensors. It requires dealing with specific protocols (OPC UA, Modbus), mapping complex data structures, and ensuring secure network configurations. The uncertainty regarding the specific SCADA vendor implementations

ID	User Story	Story Points	Justification
			warrants a higher point value.
US-16	View real-time dashboard of overall plant performance	5	Developing the initial dashboard involves both frontend UI design and backend API development. While creating basic charts is straightforward, ensuring real-time data updates and a responsive design adds moderate complexity. This story relies on the successful completion of US-01 and US-02.

Total Sprint 1 Story Points: 18

This total represents the team's commitment for the sprint, based on their historical velocity and capacity.

8. Sprint Goal and Expected Deliverables

The Sprint Goal provides a shared objective that guides the team's work during the sprint. It offers a clear focus and helps prioritize tasks if challenges arise. The expected deliverables outline the tangible outcomes that will be presented at the Sprint Review.

Sprint Goal

"Establish a reliable data ingestion pipeline for solar and wind assets and deliver a foundational real-time performance dashboard."

This goal emphasizes the critical first steps of the project: getting the data flowing and providing initial visibility to the users. It aligns with the selected user stories and sets a clear expectation for the sprint's outcome.

Expected Deliverables

At the end of Sprint 1, the team is expected to deliver the following:

1. **Functional Data Ingestion Pipeline:** A working system capable of receiving, validating, and storing real-time data from simulated solar panel sensors and wind turbine SCADA systems.
2. **Real-Time Performance Dashboard (V1):** A functional web-based dashboard displaying aggregated power output, asset status (online/offline), and basic performance metrics for the connected assets.
3. **API Documentation:** Comprehensive documentation for the newly created data ingestion and dashboard APIs.
4. **Automated Test Suite:** A suite of automated unit and integration tests covering the developed data pipelines and API endpoints.

Definition of Done (DoD)

To ensure quality and consistency, all completed user stories must meet the following Definition of Done:

- Code is written, reviewed, and merged into the main branch.
- Unit tests are written and passing with at least 80% coverage.
- Integration tests are written and passing.
- Code meets established coding standards and style guidelines.
- Feature is deployed to the staging environment.
- Feature passes User Acceptance Testing (UAT) by the Product Owner.
- Relevant documentation (API, user guides) is updated.

9. Appendix: Technical Architecture Overview

The AI-Based Smart Renewable Energy Asset Performance Monitoring System will be built on a modern, scalable, cloud-native architecture. This architecture is designed to handle high volumes of time-series data, support complex AI model inference, and provide a responsive user interface.

Data Flow

1. **Data Sources:** IoT sensors on solar panels, SCADA systems on wind turbines, battery management systems, and external weather APIs generate raw data.
2. **Ingestion Layer:** Data is securely transmitted to the cloud platform via MQTT or HTTPS. An API gateway and message broker (e.g., Apache Kafka or AWS IoT Core) handle the incoming data streams, ensuring reliable delivery and decoupling sources from processing.
3. **Processing & Storage Layer:** Stream processing engines (e.g., Apache Flink or AWS Kinesis Data Analytics) validate, clean, and normalize the data in real-time. The processed data is then stored in a high-performance time-series database (e.g., InfluxDB or TimescaleDB) for efficient querying and analysis.
4. **AI & Analytics Layer:** This layer hosts the machine learning models. It continuously queries the time-series database, performs inference (e.g., anomaly detection, failure prediction), and stores the results back into the database or triggers alerts.

5. **Application Layer:** Backend microservices (e.g., Node.js or Python/FastAPI) provide APIs for the frontend application, handling business logic, user authentication, and data retrieval.
6. **Presentation Layer:** A responsive web application (e.g., React or Angular) provides the user interface, displaying dashboards, reports, and alerts to the various stakeholders.

AI Model Selection

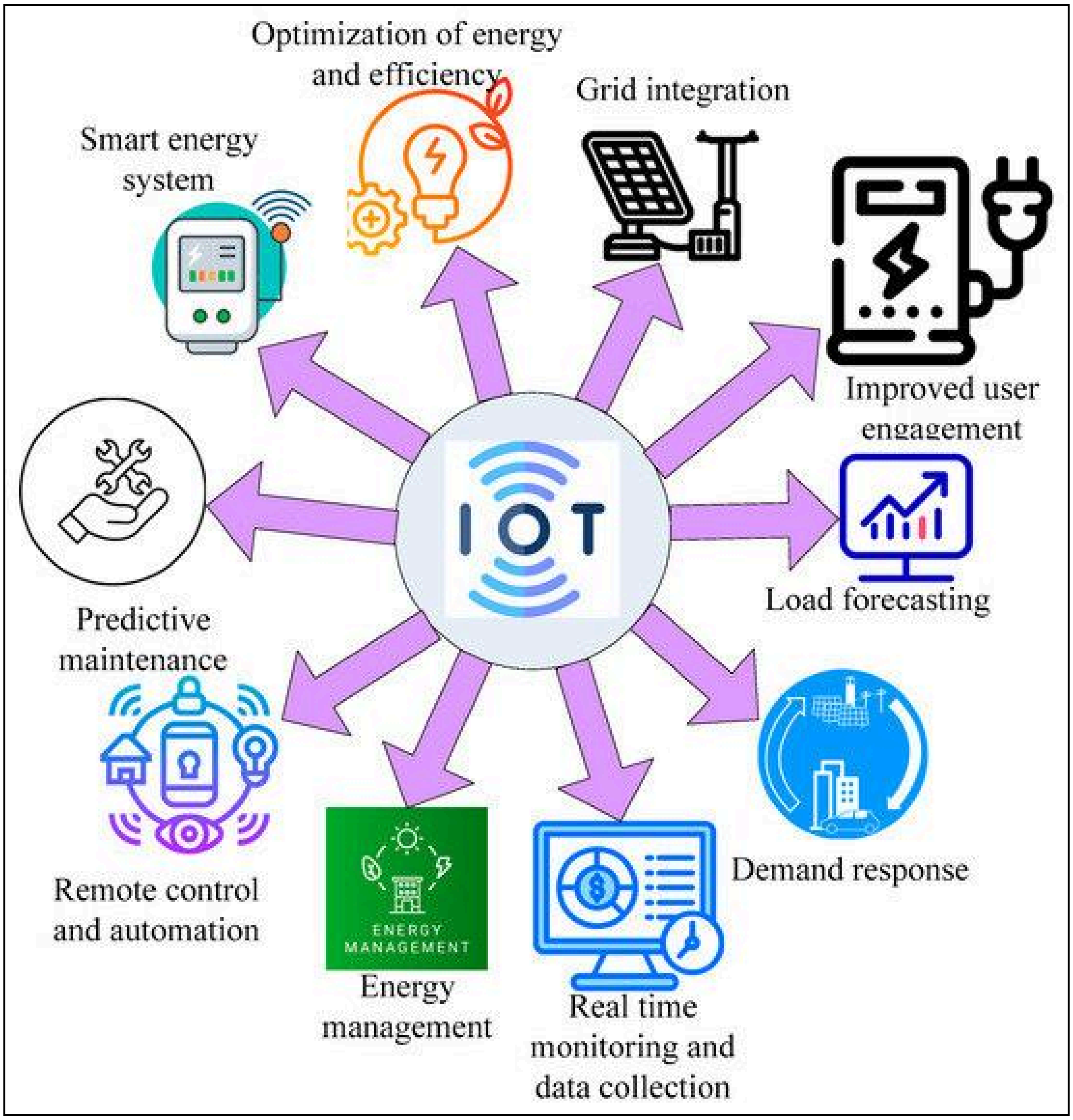
The system will utilize a variety of AI and machine learning models tailored to specific tasks:

- **Anomaly Detection:** Unsupervised learning models, such as Autoencoders or Isolation Forests, will be used to identify unusual patterns in sensor data that may indicate early-stage equipment degradation.
- **Failure Prediction:** Supervised learning models, particularly Recurrent Neural Networks (RNNs) like Long Short-Term Memory (LSTM) networks, will be employed for time-series forecasting to predict the remaining useful life (RUL) of critical components based on historical failure data.
- **Performance Optimization:** Reinforcement learning or optimization algorithms will be used to suggest operational adjustments (e.g., turbine pitch) to maximize energy output under varying environmental conditions.

Cloud Infrastructure

The platform will be hosted on a major cloud provider (e.g., AWS, Azure, or GCP) to leverage managed services for scalability, reliability, and security. Key infrastructure components will include:

- **Compute:** Containerized microservices orchestrated by Kubernetes (e.g., EKS, AKS, or GKE) for flexible and scalable application deployment.
- **Storage:** Managed time-series databases for operational data, relational databases (e.g., PostgreSQL) for user and configuration data, and object storage (e.g., S3) for drone imagery and reports.

- 

  Networking: Virtual Private Clouds (VPCs), load balancers, and API gateways to ensure secure and efficient data routing.
- **Security:** Identity and Access Management (IAM), encryption services (KMS), and security monitoring tools to protect sensitive data and infrastructure.